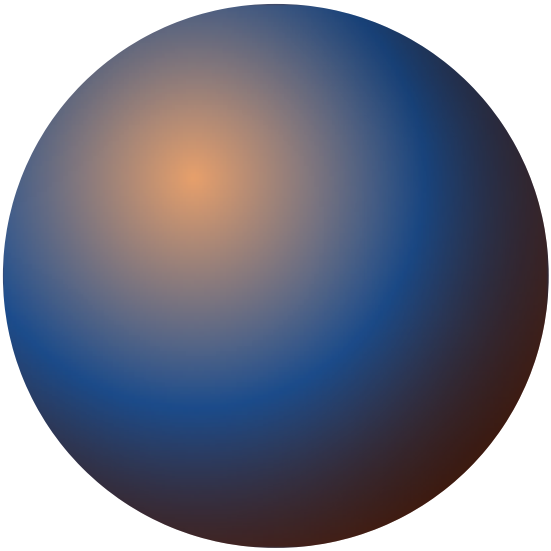
MY RESEARCH NOTEBOOK, TYPED UP · NASA SPACE APPS CHALLENGE · “INTERPLANETARY SURVIVAL GUIDE: MARTIAN MAP”

# Dr. Ayan’s Research on Mars and its Habitability

*Three versions of Mars Atlas, every formula I got wrong, and how I fixed them. Written by hand first, typed up here.*



**Author:** Ayan
**Project:** Mars Atlas
**Versions:** Omega (Ω), Alpha, Gamma
**Study site:** Jezero Crater, Mars
**Source:** my handwritten notebook
**Dedicated to:** Dr. Iroha Sakayori

# 1 Field notes

*A flimsy two chapters, finished 250 pages. Well, good luck, Future Scientist.*

NASA MARS · 10-31

Alright, the website is good to go. It works and all, however the calculations are just estimates. See, I was thinking:

Oxygen is generally calculated per hour, but that won't help for short travels. So, for oxygen on a Mars walk, i.e. for short distances, let's take:

> $O_2 = [\text{ average oxygen intake} \times 60 \text{ seconds }] + \text{reserve}$

Here, the reserve is the extra oxygen needed after accounting for fatigue, deep breaths, rests, etc. Pretty much the human calculations.

However, Mars is not a smooth sphere. It has multiple craters and what not. To solve this, or find a way around it, I'm installing an AI-guided chatbox equipped with the smartest AI model.

TWO HOURS LATER

Hi, I'm back after 2 hours. The weather is very stable on Mars, but it's usually −40 °C.

The colder the air, the lower the temperature. According to quantum mechanics, cold means less kinetic energy, and less kinetic energy means less radiation (particles do not hit each other as much). From this:

> $\text{Temperature} \propto \text{Radiation}$

Therefore, Mars would have less radiation the colder the air is! So we can calculate radiation very precisely if we have the temperature. Groundbreaking discovery!!

This applies only to thermal radiation. So using this, if we have the radioactivity, we can find the temperature, and vice versa. Although the

radiation we get through this is thermal radiation, thermal radiation is the **radiation dispersed**.

**So far I've done:**

i. Radioactivity
ii. Craters
iii. Temperature
iv. Distance
v. Oxygen
vi. Weather
vii. AI guidance

Alright, I'm now dealing with the finished product. Well, almost. I'm gonna spend a lot of time dedicated to brainstorming.

## 2 Three versions of Mars Atlas

So, to start with, I'd say I have 3 versions of Mars Atlas. I call the first version Ω, Version Omega. It taught me a ludicrous amount of stuff.

Omega attempted to answer these questions: how much oxygen will *short* travels take? And it attempted to map Mars perfectly.

By actively working on Project Omega, I discovered that multiple factors obstruct an astronaut's journey: radiation, weather, temperature.

So the next version, Alpha, aimed to address these issues.

Alpha was beautiful. It didn't teach me as much as Omega, of course, but it was successful in various aspects. It addressed radiation, temperature, craters, weather, distance, oxygen and terrain.

### The mother of all formulae

So, I developed the mother of all formulae that are in the latest version:

> ~~Radiation dispersed = Radiation × (air density)~~

During Alpha, I discovered that radiation is dispersed more the denser the air is. That means:

> less temperature = colder air = denser air = less kinetic energy = less radiation ($K.E \propto T$)

This is thermal radiation that I stumbled on. It's not the radiation dispersed; it is a major part of it.

Now, the $O_2$ required to travel across a crater:

> ~~$O_2 = \text{Time} \times [\text{ dimensions of crater} \times O_2 \text{ required to travel }]$~~

Now, you cannot calculate the amount of time needed, and you can't multiply a 3-dimensional object with any compound or element, but I will get into it later once version 3 comes.

> Total $O_2 = [\ (O_2 \text{ through craters} + O_2 \text{ anywhere else}) \times \text{crew}\ ] +$ required reserve

## Why Alpha was abandoned

These were the original formulae I created, but they lacked practical aspects. If Omega created a road for me to walk on, then Alpha would be the pathway to a successful prototype. However, it is indeed an abandoned version, for a lot of reasons.

First off, "radiation dispersed" isn't actually the correct word for that formula. It would be correct to call it thermal radiation: the radiation involving temperature. It is a part of dangerous radiation, not dispersed radiation!

Secondly, the crater oxygen calculation is just incorrect. I had to delve deep into crater geometry to properly think of a formula and research it. It is an impossible task to multiply a massive 3-dimensional figure, as it is, with the oxygen required to travel it!

The total $O_2$ formula is, however, correct. So now we have incorrect formulae, correct ideas and a whole bunch of calculations. Good. We started at just the dimensions and details of Mars. Alpha touched on a lot of topics and shed light on many more:

1. **Fatigue.** Humans get tired and start panting, especially if they walk on a completely different planet.
2. **The actual radiation dose.** Omega calculated an estimated radiation dose assuming Mars always had peak radioactivity, making the times when radioactivity is stable inaccurate. Now, Alpha calculated thermal radiation. That is an important aspect, yes, but it's not the dangerous ionizing radiation dispersed.
3. **There were no rests.** An astronaut was expected to walk at a constant speed of 4 km/h with no rest, no fatigue, no stress.

# 3 Current version: Gamma

Gamma has learnt from every single mistake made by Omega and Alpha and fixed them. It even addressed multiple unnamed issues that NASA itself hasn't.

## Revised radiation dispersed

> $\text{Dose}(X) = D_0 \times e^{-\mu X}, \text{ where } X = P \div g$
>
> P = pressure, Do = dose, g = gravity

The original radiation dispersion idea was correct; the practicality wasn't.

To get a consistency check, I measured Curiosity's radiation reading: −0.13 μGy/day per Pa. So, let's try and get this value with our formula:

$$
\begin{aligned}
dD/dP &= d/dP\,(D_0\, e^{-\mu X}) \\
&= -\mu \times (dX/dP) \times D \\
&= -0.022 \times d/dP\,(22.6\ \text{g/cm}^2) \times D \\
&= -0.022 \times 0.0270 \times D
\end{aligned}
$$

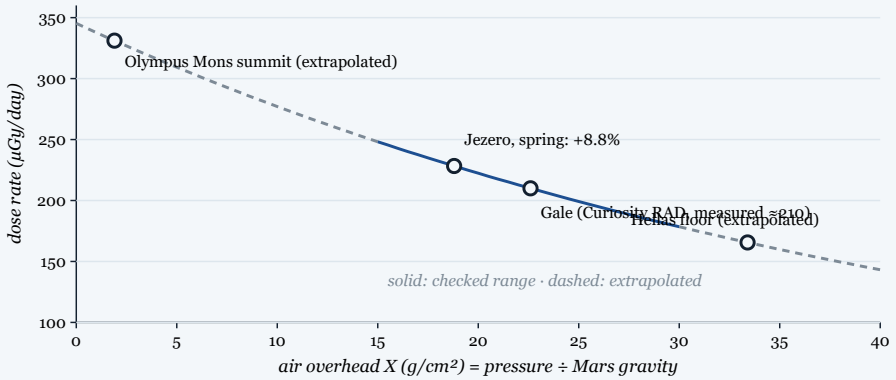
To explain dX/dP:

> $X = P \div g = P \div 3.71\ \text{kg/m}^2$
> $= (P \div 3.71) \times 1000 \div 100^2$
> $= P \div 37.1\ \text{g/cm}^2$
>
> $dX/dP = d/dP\ (P \div 37.1) = 1 \div 37.1 = 0.02695 \approx \mathbf{0.0270}$

> $dD/dP = -0.022 \times 0.0270 \times D,$ with $D = 233\ \mu\text{Gy/day}$
> $= -0.022 \times 0.0270 \times 233 = \mathbf{-0.138\ \mu Gy/day\ per\ Pa}$

That is the expected answer. The result we calculated is about 8.8% more than Gale's average. I believe that this is due to the thinner air at Jezero. Need to research more independently on this topic!

*Small graph time!!*




**Dose against air overhead.** Shows a very obvious slope. Like I said, research needed.

## My theory, revisited

> less temperature = colder air = less kinetic energy = atoms hit other atoms less = less radioactivity

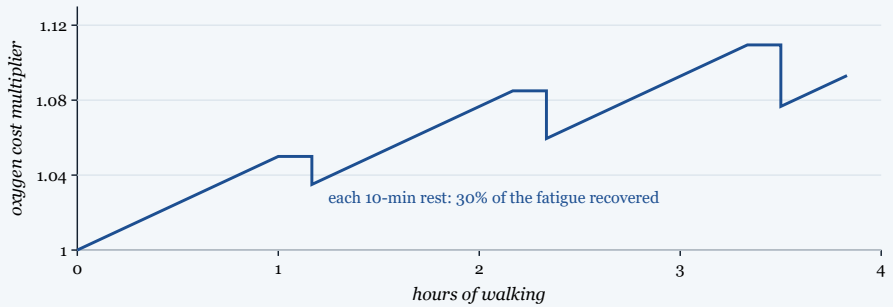
This is only partly true, for the specific instance of Mars, and Mars only!!

## Rest spots

Onto our next topic: rest spots. There's not much to say here. Humans get tired or they get stressed, so they need to rest. Not a thing one should complain about; instead, praise it. So multiple rest sites have been designed and assumed safe, due to the calculations of radiation and temperature coming out as safe for humans.

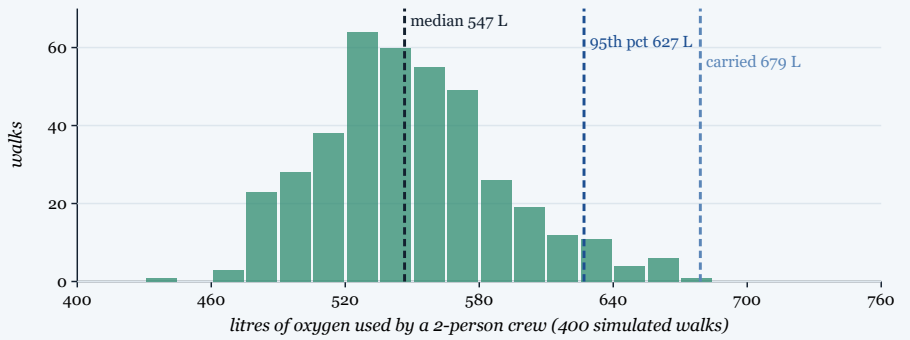
## Fatigue

Now, let's talk about fatigue. I have used multiple sources and calculated the average oxygen cost due to exhaustion. An interesting thing to note is that it increases per hour walked.




**Oxygen cost multiplier against hours of walking.**

Stress breathing is highly unpredictable, so I observed multiple journeys and noticed the lows, peaks and average:




**Litres of O2 used by a 2-person crew, 400 walks.** Median 547 L.

So Alpha's O2 calculation was in fact correct:

> Total $O_2 = [\,(O_2 \text{ through craters} + O_2 \text{ everywhere else}) \times \text{crew}\,] + \text{reserve}$

## 4 Craters

Now let's talk about the hardest part, which I saved for last.

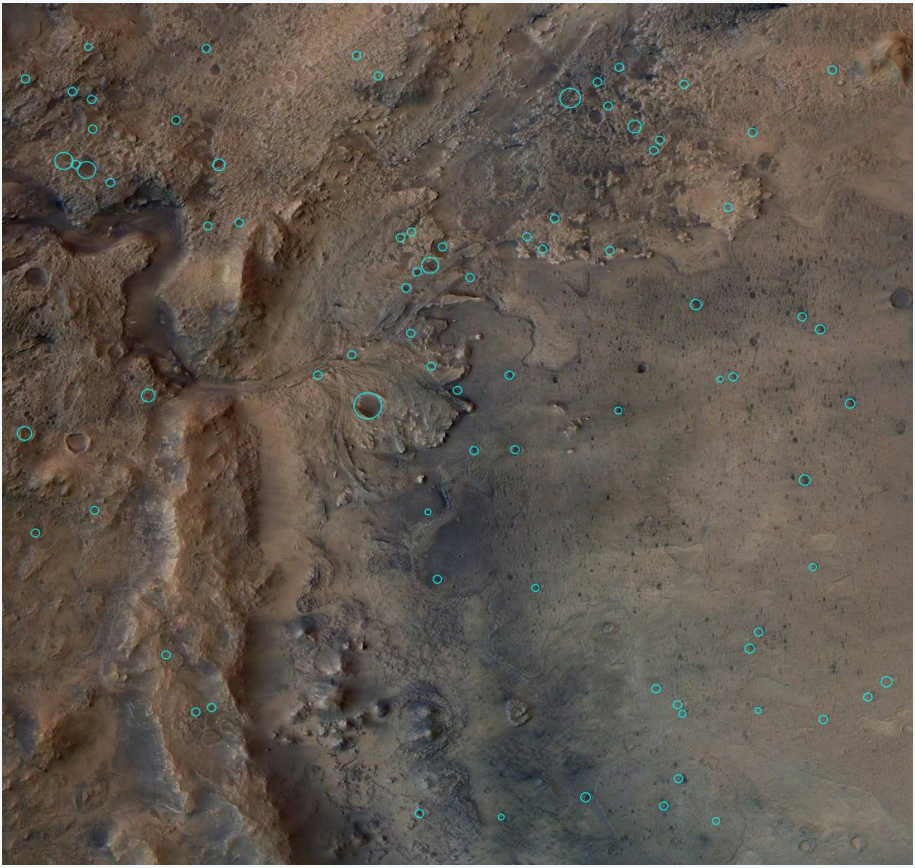
Craters, out of everything in the research project, might just matter the most. They single-handedly decide how much more oxygen will be required, how much dosage of radiation the astronaut will be exposed to, and how long the journey may be.

The detector works on the DEM at 40 m spacing. Candidate centres are the low (deep) points. Around each of them there are 24 directions at growing radii. It then averages the elevation around the circle at each radius, which cancels noise and leaves a clean bowl. Tests for roundness and depth remove false hits.

I failed once: Belva measured 1,410 m, which is very far off the correct number.

But still, to combat this without switching to AI right away, I've calculated multiple craters. Actually, no: I have identified every single crater on Mars

and calculated their elevations with the help of AI!



**The craters found around Jezero.** The big ring on the delta is Belva.

These craters will be a very significant issue to battle. However, not the *most* significant.

## Radiation

Mars is not a habitable planet. It may have been habitable eons ago, but in its current state, it's dead.

| Lowest radiation recorded | |
|---|---|
| Surface average | 0.64 mSv/day |
| Hellas Planitia | 0.342 mSv/day |
| Lava tube SW of Hadriacus Mons | 0.064 mSv/day, comparable to Earth! |

However, these are the lowest radiation levels recorded in history.

| Highest radiation recorded | |
|---|---|
| Surface | 8,100 micrograys, equal to 30 chest X-rays |
| Average overall surface | 8.8 micrograys/hour |

These numbers don't sound like much, do they? Alright then, here's the sweet spot of micrograys on Earth: the world average is 2.4 mSv/year.

Mars has tens to hundreds of times higher radiation dose than Earth in specific places! Astronauts need to be very wary of radiation, craters, stamina, etc.

## Weather

But we haven't even touched the most sensitive, least predictable topic: weather. How do we even predict Mars' weather? It is so very confusing and fascinating. I will come back after some research. But from what I know, Mars has very extreme temperature fluctuations. Apparently it can go from −153 °C to −20 °C. That is enormous. But it follows a seasonal pattern. #science

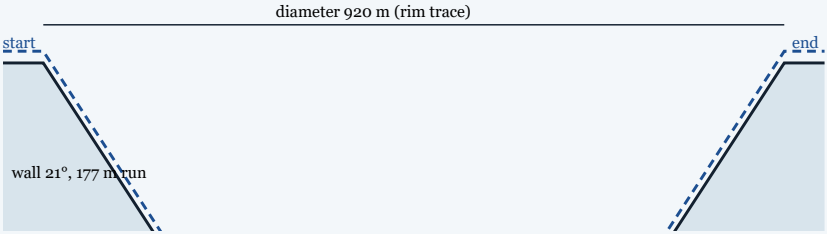
## Testing on Belva

I turned the tests stricter, so that they only left 1 crater, then 4 craters. Then I switched to averaging the circle profile, which fixed it: **81 craters**, 76 under 500 m and 5 between 500 and 1,000 m.

I then decided to leave every other crater alone and focus only on Belva. Belva has a published width of 900 m (0.9 km), so let's test!

Elevation data: 1 cell = 40 m
Depth = median crest height − lowest point (within 200 m of centre) = **68 m**
Wall slope = **21°**
Depth ÷ diameter = 68 ÷ 920 = **0.074**

tan 21° = 0.3839
Wall run = depth ÷ tan θ = 68 ÷ 0.3839 = **177.2 m**
Wall grade = depth ÷ wall run = 68 ÷ 177.2 = 0.384 = **38.4%**
Floor = 920 − (2 × 177.2) = 920 − 354.4 = **565.7 m**
Approach and exit = 50 m each
Across = 50 + 177.2 + 565.7 + 177.2 + 50 = **1,020 m**
Around = π × r,  r = 920 ÷ 2 + 50 = 460 + 50 = 510 m
∴ Around = 3.14 × 510 = **1,602 m**




**Belva in cross-section**, with the path straight across.

Detour = 1,602 − 1,020 = **582 m**

Detour is the extra distance you travel to go around.

Good. We got 920 m and the proper measurements for our journey, with a few angles. If you thought this was hard, think again: we now calculate the oxygen required.

## Oxygen across Belva

We're taking $VO_2$, i.e. oxygen depending on velocity:

> Flat = 14.04 · Down the wall = 15.48 · Up the wall = 15.48 (ml/kg/min)

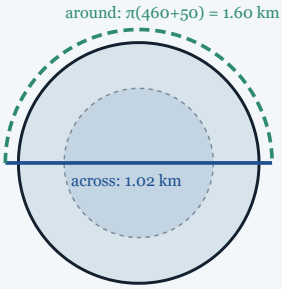
I will spare you the tedious calculations, but if you wish to verify, the formula is: 43% effort, $VO_2 = 0.43 \times 36$, which stays constant.

### Time and $O_2$ for one person:

> Litres = $VO_2 \times 82\ \text{kg} \times 60\ \text{min/h} \times \text{hours} \div 1000 = VO_2 \times 4.92 \times \text{hours}$

| Piece | Time | $O_2$ |
|---|---|---|
| Approach + exit (50 + 50 = 100 m) | 0.1 ÷ 2.74 = 0.0365 h | 14.04 × 4.92 × 0.0365 = 2.52 L |
| Down the wall (177 m) | 0.1772 ÷ 1.85 = 0.0959 h | 15.48 × 4.92 × 0.0959 = 7.31 L |
| Floor (566 m) | 0.5657 ÷ 2.74 = 0.2065 h | 14.04 × 4.92 × 0.2065 = 14.26 L |
| Up the wall (177 m) | 0.1772 ÷ 0.87 = 0.2027 h | 15.48 × 4.92 × 0.2027 = 15.44 L |
| **Across, total** | **0.542 h = 32.5 min** | **39.5 L** |
| **Around (1,602 m)** | **1.602 ÷ 2.74 = 0.585 h = 35.1 min** | **14.04 × 4.92 × 0.585 = 40.4 L** |

For 2 persons: about **80 L** across, **81 L** around.




Belva, 2 crew, 43% effort:

across 80 L (69–93 L) 33 min

around 81 L (70–93 L) 35 min

Detour 582 m, but no 68 m climb.
Walls 21° > 20° hazard limit,
so the site recommends going around.

Ranges: 90% of 300 runs with diameter ±5%, depth ±15%, wall ±3°, people ±10%.

**Across against around.**

## The cherry on top for Gamma: Atlas

An astronaut AI specifically trained and engineered to guide astronauts and do all the calculations about 300 times before giving you a proper answer. All done in 1–2 seconds.

# Version Gluon

How consistent are marsquakes? How catastrophic are they? I’ve measured dangerous radiation, and everything, but now’s the time for “natural” events.

My notes quote around 10/day during Martian summer (2025 - New Scientist), and ~2 per day by the end of 2019. That is a big difference, but I still need to check the original sources. InSight detections are not a complete global daily count.

Can we predict this? I’ll check it out.

Okay, many detected marsquakes are weaker than major earthquakes, but that alone does not mean we can ignore seismic risk.

Next topic - thermal radiation.

Mars’ surface pressure is roughly 1% of Earth’s. That is not a measure of atmospheric thickness or a 99:1 rule for radiation shielding. The column mass, particle type and energy matter here.

# Thermal radiation

HEAT BALANCE

Let’s calculate the dangerous amount of thermal radiation an astronaut can be exposed to.

Let’s first optimize the heat.

Heat in = Heat out, if the body and suit are in steady heat balance.

Assume body core temp: 37 °C. For radiation from the suit, though, I need the outer surface temperature, not the core temperature.

Heat in = Metabolic heat + absorbed sunlight

Heat out includes thermal radiation, suit cooling, evaporation and other heat-transfer paths.

For a rough starting assumption: metabolic heat ≈ 100 W at rest and 300 W during work. These change with activity.

# Temperature and radiation

THE QUESTION I AM TESTING

Finding a relation between temperature and radiation for the specific instance of Mars.

Resources used: Curiosity, NASA, Claude research agent, Space.com.

My notes record a low of −80 °C on 29-01-2022. I still need to verify the date and context against the original source.

My notes also record a dose rate up to 3.5× normal on 16 Feb 2022. This needs its original source checked too.

I am not going to assume any connection.

Onto the next data.

Mars Atlas: $\text{Dose}(X) = D_0 \times e^{-\mu X}$
$X = P / g$
$\mu = 0.022\ \text{cm}^2/\text{g}$
This is a simplified attenuation model, not a validated description of the whole Martian radiation field. X and μ need consistent units.

For an ideal gas, mean molecular translational kinetic energy is proportional to absolute temperature. That does not mean ambient temperature sets the ionizing dose.

What is radiation exactly?

# What is radiation exactly?

## THERMAL AND IONIZING RADIATION

Thermal radiation → a temperature-dependent spectrum, mainly infrared at ordinary environmental temperatures.

Ionizing radiation → enough energy per particle or photon to ionize atoms or molecules. Cosmic rays and radioactive decay can produce it; it can damage cells and DNA.

So, colder air means less average molecular kinetic energy. But fewer thermal collisions do not directly remove the energetic particles RAD measures. That step in my reasoning needs to stay a question, not a conclusion.

Alright, colder air = denser air holds for an ideal gas if pressure and composition stay constant. Atmospheric shielding depends on column mass, approximately P/g, rather than local density alone.

Let’s see what RAD dosage says & REMS weather says.

Wow (65 + 80 + 22) million characters. That took quite the time. But now I have what I truly need.

# Timing first

OBSERVATION 00

Juicy part first.

| Timing check | RAD | REMS / derived |
|---|---|---|
| UTC at RAD start | 2019-09-26 08:25:42 | Same target instant |
| Mars clock at that instant | 2538 00:15:58 | 2538 00:02:51 (estimated) |
| Clock deviation | RAD is ahead | ~13 min 7 sec behind |

Observation 00 is essentially defining our data & the deviation we will have.

I started this research with very little expectations - suffice to say I am now delighted.

UTC gap: REMS sample − RAD start

REMS UTC is estimated from the label endpoints and table LMST. The ~13:07 difference is a Mars-clock discrepancy at the same estimated instant, not the nearest-sample UTC gap. It varies to about 13:17. I have not applied a validated clock correction.

# Highest and lowest readings

RAD AND REMS · DETECTOR E

| Reading | Obs. | E dose µGy/h | Nearest air K | Mean air K / °C |
|---|---|---|---|---|
| Highest | 90 | 15.08608 | 222.47 | 228.40 / −44.75 |
| Lowest* | 88 | 8.15563 | 223.68 | 223.12 / −50.03 |
| Lowest RAD QC-zero | 16 | 12.36053 | 199.13 | No samples |
| Lowest QC-zero, ≥90% sampled | 36 | 12.44899 | 194.14 | 193.95 / −79.20 |

Observations 88 & 90, when looked at a glance, differ by only 1.21 K in their nearest recorded temperatures. Averaging the samples over their estimated observation windows gives 90 → 228.40 K and 88 → 223.12 K.

That leaves a 5.27 K difference between the means. It caught my attention, but I cannot say it is too large to be a coincidence. The larger 34.45 K difference compares 90 with 36 instead, so it is a different pair.

# Now, for the quality check

THE QUESTION IS STILL OPEN

I’m not gonna write all the 80 readings they give in my PDF. Only those who matter will be involved.

Obs. 88 unfortunately has RAD QC 00000001, so it is flagged. Obs. 16 is the lowest dose with zero RAD QC, but has no temperature sample inside its estimated observation window.

The nearest reading for Obs. 16 is about eight minutes earlier. Obs. 36 has 902 temperature samples over 961 seconds: nearest air 194.14 K, mean 193.95 K. Obs. 90 has 961/961 samples, and 88 has 649/649. These are sampled window averages, not temperatures inferred at an exact radiation site.

This leaves my hypothesis open, which makes me happy. It does not prove a relationship or establish sensible grounds by itself, and it is not enough to prove the hypothesis correct.

Next step: compare all 22 observations with zero RAD QC and at least 90% temperature sampling under the approximate UTC mapping. No correlation test has been completed yet.

DEDICATION

# For Dr. Iroha Sakayori

This notebook is dedicated to Dr. Iroha Sakayori of *Cosmic Princess Kaguya!*, my inspiration.

She is fictional, but what she stands for is real to me. She started as an ordinary student juggling school and work, chose a goal so big it needed computer science, robotics, neuroscience and more, and spent years learning all of them until she earned her doctorate.

I study for the same reason she did: because the universe is too fascinating not to. Every page in here, every wrong formula and every fixed one, is a small step toward becoming a scientist like her.

*Thank you, Dr. Sakayori. — Ayan*